

Figure 2A presents the average annual citation trajectories of three reviewer invitee groups: Accept (gray), Click-Denial (red), and Auto-Denial (blue). Prior to the review invitation, citation patterns for the Click-Denial and Auto-Denial groups are nearly identical (Wald test: $p = 0.543$), supporting the comparability of these two non-reviewing groups. In contrast, the Accept group displays a markedly higher baseline citation rate to the corresponding author, vividly illustrating the self-selection challenge: reviewers who agree to evaluate a manuscript are systematically more familiar with or interested in the authors' work, biasing post-review comparisons. Consequently, to ensure valid causal inference, our analysis strategically focuses on comparing the Click-Denial and Auto-Denial groups, thereby isolating incidental exposure effects from reviewer self-selection.

While the citation trajectories for the Click-Denial and Auto-Denial groups were indistinguishable prior to the review invitation, they diverge notably afterward (Figure 2A). Reviewer invitees who actively clicked to decline—thus likely exposed to the manuscript's metadata—show a sustained increase in citations to the corresponding author's work. In contrast, the Auto-Denial group, who passively ignored the invitation, maintains a flat trajectory. Five years after the invitation, Click-Denial reviewer invitees cite the corresponding author's work 29.09% more than the Auto-Denial group. These results suggest that concise exposure to a manuscript's core ideas during the invitation process can meaningfully and persistently influence the diffusion of scientific knowledge, despite reviewer invitees not accessing the full manuscript during the process.

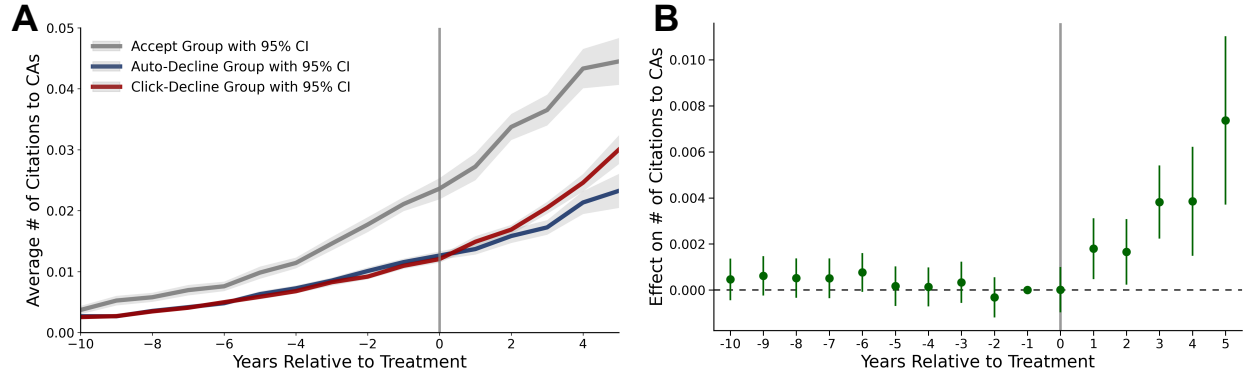


Figure 2. Citations to corresponding authors over time and event-study estimates. (A) Average number of citations from reviewer invitees to the corresponding authors (CAs) of manuscripts they were invited to review, plotted by year relative to the invitation (Year 0). The plot compares the Accept group, Click-Delay (treatment) group and Auto-Delay (control) group. Shaded areas represent 95% confidence intervals. **(B)** Event-study estimates from the regression, restricted to the Click-Delay and Auto-Delay groups. The model includes relative-year indicators interacted with treatment status to estimate the dynamic effect of minimal exposure on subsequent citations to the corresponding authors.

To systematically test the observed effect controlling for moderators [19, 20], we employ a DiD regression model on the Click-Delay and Auto-Delay subsample:

$$y_{it} = \beta_0 Treat_i \times Post_t + \beta_1 Treat_i + \beta_2 Post_t + \beta_3 C_i + \epsilon_{it}$$

where y_{it} denotes the number of citations from reviewer invitee i to the corresponding author in year t ; $Treat_i$ is an indicator for membership in the Click-Delay group; and $Post_t$ marks years after the invitation. The interaction term, $Treat_i \times Post_t$, captures the differential change in citations for the treatment group post-invitation. Figure 2B reports event-study estimates, confirming parallel pre-trends and revealing a statistically significant post-invitation increase in citation behavior for the Click-Delay group.

Model (1) in Table 1 reports the baseline model estimates: Click-Delay group cites corresponding authors 0.0021 more times than Auto-Delay group after the invitation ($p = 0.000$). While modest in absolute terms, this reflects a 14% relative increase compared to the Auto-Delay group's post-invitation citation rate. Event study estimates reveal that the increase begins in the invitation year and persists over time. To account for potential confounders, we

extend the DiD model with covariates in Model (2) capturing: (1) author–reviewer proximity (prior co-authorships and mutual citations), to address pre-existing intellectual connections before invitation; (2) reviewer invitees’ experience (publication count and *h*-index), to control for seniority and scholarly productivity before invitation; and (3) engagement with the journal (past publications in and citations to the journal), to capture alignment with the journal’s content before invitation. See statistical descriptions of these variables in SI S2.

Including these controls modestly changed the treatment effect to 0.0018 ($p = 0.000$), corresponding to a 12% relative increase. This reinforces that the observed differences stem from exposure to the invitation rather than from individual-specific characteristics. We further validate our findings using Two-Way Fixed Effects (TWFE) and Staggered DiD models [21], which address treatment timing heterogeneity and potential confounding from staggered exposures. These specifications yield consistent estimates, detailed in SI S3, confirming the robustness of the effect.

These findings support the hypothesis that peer review facilitates the diffusion of new and related scientific ideas. While exposure in this setting typically involves only a brief inspection of a manuscript’s title and abstract, the magnitude and persistence of its effect become substantial when aggregated across the millions of peer review invitations extended annually.

TABLE 1: BASELINE ESTIMATES

	(1)	(2)
$Treat_i \times Post_t$	0.0021*** (0.0004)	0.0018*** (0.0004)
$Treat_i$	-0.0003* (0.0002)	0.0003* (0.0001)
$Post_t$	0.0087*** (0.0003)	0.0079*** (0.0003)
# Collaboration for Both		-0.0059 (0.0037)
# Citation Reviewer to Author		0.0060*** (0.0004)
# Citation Author to Reviewer		-0.0004* (0.0002)
Reviewer # Publications (log)		-0.0035*** (0.0001)
Reviewer H-index (log)		0.0048*** (0.0002)
Reviewer # Pub on Journal		-0.0001** (0.0000)
Reviewer # Ref to Journal		-0.0000* (0.0000)
Const	0.0065*** (0.0001)	0.0079*** (0.0008)
Observations	4,755,571	4,755,571
No. clusters	450,737	450,737
R-squared	0.0014	0.1272

Table 1. The effect of exposure via review invitation on downstream citations to corresponding authors.

This table presents regression estimates using a panel dataset of reviewer–manuscript pairs spanning ten years before to five years after the review invitation. The dependent variable is the average annual number of citations made by the reviewer invitees to the corresponding author. Model (1) shows that reviewer invitees who Click-Declined the invitation cite corresponding authors 0.0021 more times per year after the invitation than those who Auto-Declined ($p = 0.000$). This effect represents a 14% relative increase compared to the post-invitation citation rate of the Auto-Delay group. Model (2) adds covariates controlling for prior connections to the corresponding author (co-authorships and mutual citations), reviewer invitees' experience (career H-index, total publications), and prior engagement with the journal. The estimated effect remains statistically significant at 0.0018 citations per year ($p = 0.000$), reflecting a 12% relative increase. Robust standard errors clustered at the reviewer invitee level are reported in parentheses (** $p < 0.01$, * $p < 0.05$, * $p < 0.1$).

Causal Identification

To further strengthen the causal interpretation of our findings, we implement an instrumental variable approach using a natural experiment that induces exogenous variation in exposure to peer review invitations [18, 22]. Specifically, we exploit the quasi-random variation in the timing of invitation emails—comparing those sent on weekdays (Monday through Friday) to those sent on weekends (Saturday and Sunday). Of the 551K total invitations in our dataset, 90.06% (496K) were issued on weekdays, whereas 9.94% (55K) were sent on weekends.

This timing-based variation serves as a plausibly exogenous shock to exposure likelihood. Given that scientists systematically exhibit reduced email activity during weekends, invitations sent on weekends experience significantly lower response rates. Consequently, weekend timing serves as a natural instrument for exposure to manuscript content. Indeed, because the day on which ScholarOne dispatches a review invitation is unrelated to reviewer-specific factors such as research interests, prior familiarity with the manuscript, or future citation intentions, our instrumental variable approach satisfies the exclusion restriction, as illustrated conceptually in Figure 3.

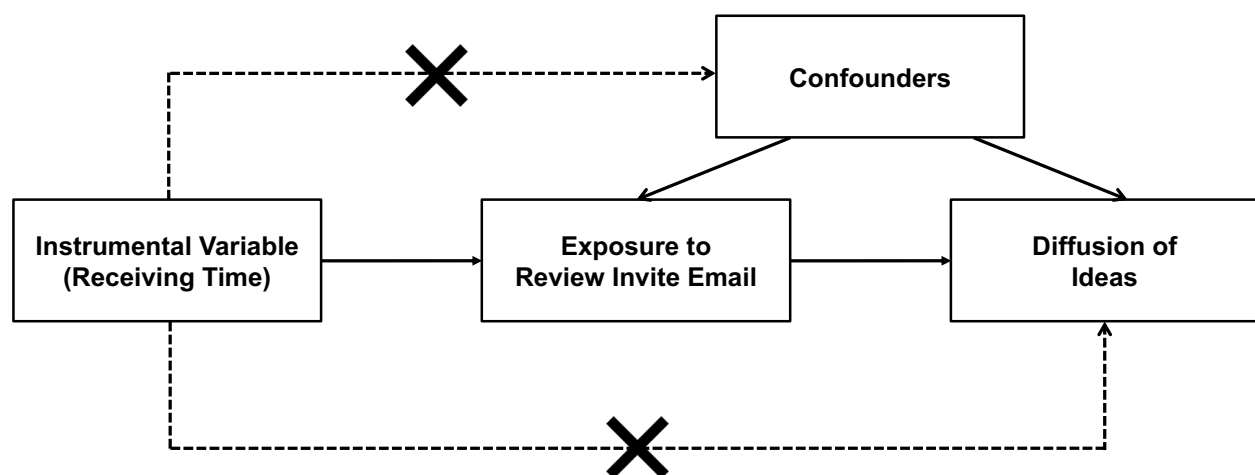


Figure 3. Instrument Variable Framework. The diagram illustrates the identification strategy: the instrument influences adoption of related ideas solely through its effect on exposure to the review invitation. It is assumed to be independent of unobserved confounders and to have no direct effect on the outcome, as denoted by the dashed arrows crossed out. This structure enables causal inference by isolating exogenous variation in exposure.